Note: When doing any first law analysis, always define the control volume with a sketch!

5-28E

5–28E The stators in a gas turbine are designed to increase the kinetic energy of the gas passing through them adiabatically. Air enters a set of these nozzles at 300 psia and 700°F with a velocity of 80 ft/s and exits at 250 psia and 645°F. Calculate the velocity at the exit of the nozzles.

Given:

- State 1 (entering nozzle)
  - $P_1 = 300$ psia, $T_1 = 700°\text{F}$, $v_1 = 80$ ft/s
- State 2 (exiting nozzle)
  - $P_2 = 250$ psia, $T_2 = 645°\text{F}$
- Adiabatic process (no heat transfer, $\dot{Q} = 0$)

Find:

- Velocity at state 2, v_2

Writing the first law of thermodynamics for an open system (nozzle):

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left[h_i + \frac{1}{2}v_i^2 + gz_i\right] - \sum \dot{m}_e \left[h_e + \frac{1}{2}v_e^2 + gz_e\right]$$

We can simplify the first law eqn for this system as follows:

- No work is done by the system ($\dot{W} = 0$), it only increases the velocity of the gas
- No heat transfer ($\dot{Q} = 0$)
- Steady state ($\frac{dE}{dt} = 0$), because there is no accumulation of energy in the system over time
- Neglect potential energy changes ($gz_i \approx gz_e$)

Thus, the first law simplifies to:

$$\dot{m}_1 \left(h_1 + \frac{1}{2}v_1^2\right) = \dot{m}_2 \left(h_2 + \frac{1}{2}v_2^2\right)$$

Applying the conservation of mass:

$$\frac{dm}{dt} = \sum \dot{m}_i - \sum \dot{m}_e$$

$$\implies \dot{m}_1 = \dot{m}_2$$

Therefore:

$$h_1 + \frac{1}{2}v_1^2 = h_2 + \frac{1}{2}v_2^2$$

Now there are two ways to determine the specific enthalpies h_1 and h_2:

Method 1: Using Property Tables

Using table A-17E (ideal gas properties for air), we can find the specific enthalpies at the given states:

- At state 1 ($P_1 = 300$ psia, $T_1 = 700°\text{F} = 1159.67\,\text{R}$):
  - $h_1 = 281.14$ Btu/lbm
- At state 2 ($P_2 = 250$ psia, $T_2 = 645°\text{F} = 1104.67\,\text{R}$):
  - Using linear interpolation between the values at $T = 1080\,\text{R}$ and $T = 1120\,\text{R}$:

$$h - h_{1080} = \frac{h_{1120} - h_{1080}}{T_{1120} - T_{1080}}(T - T_{1080})$$

$$h_2 = 267.17 \text{ Btu/lbm}$$

Substituting the values into the first law equation:

$$v_2 = \sqrt{2(h_1 - h_2) + v_1^2}$$

$$v_2 = \sqrt{2\left[281.14 - 267.17\,\frac{\cancel{\text{Btu}}}{\cancel{\text{lbm}}} \cdot \frac{778.169\,\text{ft}\cdot\cancel{\text{lbf}}}{1\,\cancel{\text{Btu}}} \cdot \frac{32.174\,\frac{\cancel{\text{lbm}}\cdot\text{ft}}{\text{s}^2}}{1\,\cancel{\text{lbf}}}\right] + \left[80\,\frac{\text{ft}}{\text{s}}\right]^2} = \boxed{840.195\,\frac{\text{ft}}{\text{s}}}$$

Method 2: Using Ideal Gas Relations

Assuming that air behaves as an ideal gas and that the specific heats are constant (we can usually assume this unless stated otherwise), we can use the following relation:

$$(h_2 - h_1) = c_p(T_2 - T_1)$$

For air at 700°F, $c_p = 0.254$ Btu/lbm · R, so:

$$h_1 - h_2 = -c_p(T_2 - T_1) = -\left[0.254\,\frac{\text{Btu}}{\text{lbm}\cdot\text{R}}\right][1104.67 - 1159.67\,\text{R}] = 13.97\,\frac{\text{Btu}}{\text{lbm}}$$

Substituting into the first law equation:

$$v_2 = \sqrt{2\left[13.97\,\frac{\cancel{\text{Btu}}}{\cancel{\text{lbm}}} \cdot \frac{778.169\,\text{ft}\cdot\cancel{\text{lbf}}}{1\,\cancel{\text{Btu}}} \cdot \frac{32.174\,\frac{\cancel{\text{lbm}}\cdot\text{ft}}{\text{s}^2}}{1\,\cancel{\text{lbf}}}\right] + \left[80\,\frac{\text{ft}}{\text{s}}\right]^2} = 840.195\,\frac{\text{ft}}{\text{s}}$$

Note that the specific heat values are dependent on temperature, so if we used the specific heat at the wrong temperature, we would see a larger error.

5-47

5–47 Refrigerant-134a enters a compressor at 180 kPa as a saturated vapor with a flow rate of 0.35 m³/min and leaves at 900 kPa. The power supplied to the refrigerant during the compression process is 2.35 kW. What is the temperature of R-134a at the exit of the compressor? *Answer:* 52.5°C

Given:

- State 1 (compressor inlet)
  - Saturated vapor at $P_1 = 180$ kPa, $\dot{v}_1 = 0.35\ \text{m}^3/\text{min}$
  - Because it is a saturated vapor, we can find its properties from the property tables:
    * $T_1 = T_{\text{sat}} = -12.73°\text{C}$
    * $h_1 = h_g = 242.90$ kJ/kg
    * $\cancel{v}_1 = \cancel{v}_g = 0.11049\ \text{m}^3/\text{kg}$
- State 2 (compressor outlet)
  - $P_2 = 900$ kPa
- Power supplied to compressor, $\dot{W} = 2.35$ kW

Find:

- Temperature at state 2, T_2

Writing the first law of thermodynamics for an open system (compressor):

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left[h_i + \frac{1}{2}v_i^2 + gz_i\right] - \sum \dot{m}_e \left[h_e + \frac{1}{2}v_e^2 + gz_e\right]$$

We can simplify the first law eqn for this system as follows:

- No heat transfer ($\dot{Q} = 0$)
- Steady state ($\frac{dE}{dt} = 0$), because there is no accumulation of energy in the system over time
- Neglect potential energy changes ($gz_i \approx gz_e$)
- Neglect kinetic energy changes ($\frac{1}{2}v_i^2 \approx \frac{1}{2}v_e^2$), because the compressor primarily ONLY increases the pressure, not the velocity

Thus, the first law simplifies to:

$$0 = -\dot{W} + \dot{m}_1 h_1 - \dot{m}_2 h_2$$

Applying the conservation of mass:

$$\frac{dm}{dt} = \sum \dot{m}_i - \sum \dot{m}_e \quad \Longrightarrow \quad \dot{m}_1 = \dot{m}_2$$

Therefore:

$$\dot{W} = \dot{m}_1(h_1 - h_2)$$

Solve for the mass flow rate $\dot{m}_1$:

$$\dot{m} = \rho\cancel{\dot{V}} = \frac{\cancel{\dot{V}}}{\cancel{v}}$$

$$\dot{m}_1 = \frac{\left[0.35\,\frac{\mathrm{m}^3}{\mathrm{min}} \cdot \frac{1\,\mathrm{min}}{60\,\mathrm{s}}\right]}{\left[0.11049\,\frac{\mathrm{m}^3}{\mathrm{kg}}\right]} = 0.05279\,\frac{\mathrm{kg}}{\mathrm{s}}$$

Plugging in the values to solve for h_2:

$$h_2 = h_1 - \frac{\dot{W}}{\dot{m}_1} = \left[242.90\,\frac{\mathrm{kJ}}{\mathrm{kg}}\right] - \frac{\left[-2.35\,\mathrm{kW} \cdot \frac{1\,\frac{\mathrm{kJ}}{\mathrm{s}}}{1\,\mathrm{kW}}\right]}{\left[0.05279\,\frac{\mathrm{kg}}{\mathrm{s}}\right]} = 287.416\,\frac{\mathrm{kJ}}{\mathrm{kg}}$$

<u>Note:</u> Energy IN is always negative by our sign convention!

Based on h_2 we can determine the state of the refrigerant at state 2. At $P_2 = 900$ kPa, $h_g = 269.31$ kJ/kg, therefore the refrigerant is a *superheated vapor*. Based on the superheated vapor tables, we can then find the temperature at state 2 using linear interpolation between the values at ($T = 50°$C, $h = 284.79$ kJ/kg) and ($T = 60°$C, $h = 295.15$ kJ/kg):

$$T - T_{50} = \frac{T_{60} - T_{50}}{h_{60} - h_{50}}(h - h_{50})$$

$$\boxed{T_2 = 52.53°\mathrm{C}}$$

5-57

5–57 A portion of the steam passing through a steam turbine is sometimes removed for the purposes of feedwater heating as shown in Fig. P5–57. Consider an adiabatic steam turbine with 12.5 MPa and 550°C steam entering at a rate of 20 kg/s. Steam is bled from this turbine at 1000 kPa and 200°C with a mass flow rate of 1 kg/s. The remaining steam leaves the turbine at 100 kPa and 100°C. Determine the power produced by this turbine. *Answer:* 15,860 kW

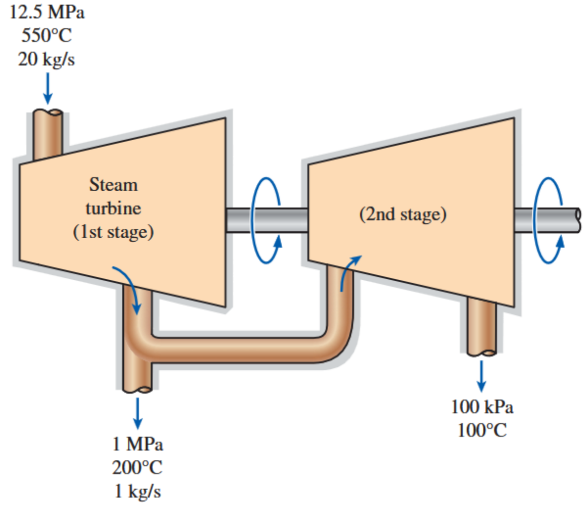


FIGURE P5–57

Given:

- State 1 (entering 1st stage)
  - $P_1 = 12.5$ MPa, $T_1 = 550°\text{C}$, $\dot{m}_1 = 20$ kg/s
- State 2 (exiting 1st stage)
  - $P_2 = 1$ MPa, $T_2 = 200°\text{C}$, $\dot{m}_2 = 1$ kg/s
- State 3 (entering 2nd stage)
  - $P_3 = 1$ MPa, $T_3 = 200°\text{C}$, $\dot{m}_3 = 20 - 1 = 19$ kg/s
- State 4 (exiting 2nd stage)
  - $P_4 = 0.1$ MPa, $T_4 = 100°\text{C}$, $\dot{m}_4 = \dot{m}_3 = 19$ kg/s
- Adiabatic process (no heat transfer, $\dot{Q} = 0$) with STEAM

Find:

- Power produced by the turbine, $\dot{W}$

Writing the first law of thermodynamics for an open system (both stages of the turbine):

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left[h_i + \frac{1}{2}v_i^2 + gz_i\right] - \sum \dot{m}_e \left[h_e + \frac{1}{2}v_e^2 + gz_e\right]$$

Since the turbine's main purpose is to convert the enthalpy of the steam into work, we can simplify the first law equation as follows:

- No heat transfer ($\dot{Q} = 0$)

- Steady state ($\frac{dE}{dt} = 0$), because there is no accumulation of energy in the system over time
- Neglect potential energy changes ($gz_i \approx gz_e$)
- Neglect kinetic energy changes ($\frac{1}{2}v_i^2 \approx \frac{1}{2}v_e^2$), because the compressor primarily ONLY increases the pressure, not the velocity

Thus, the first law simplifies to:

$$\dot{W} = [\dot{m}_1 h_1] - [\dot{m}_2 h_2 + \dot{m}_4 h_4]$$

Using the superheated water tables, we can find the specific enthalpies: $h_1 = 3476.5$ kJ/kg, $h_2 = 2828.3$ kJ/kg, and $h_4 = 2675.8$ kJ/kg.

Substituting into the first law equation:

$$\dot{W} = \left[20\,\frac{\text{kg}}{\text{s}} \cdot 3476.5\,\frac{\text{kJ}}{\text{kg}}\right] - \left[1\,\frac{\text{kg}}{\text{s}} \cdot 2828.3\,\frac{\text{kJ}}{\text{kg}} + 19\,\frac{\text{kg}}{\text{s}} \cdot 2675.8\,\frac{\text{kJ}}{\text{kg}}\right] = \boxed{15861.5\,\frac{\text{kJ}}{\text{s}}}$$

5-63

5–63 A saturated liquid–vapor mixture of water, called wet steam, in a steam line at 1500 kPa is throttled to 50 kPa and 100°C. What is the quality in the steam line? *Answer:* 0.944

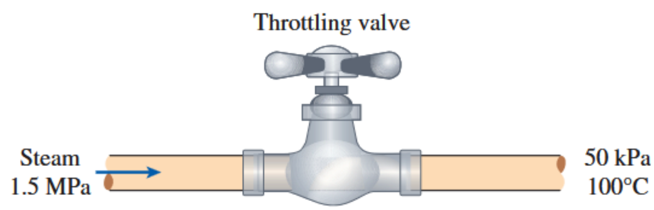


FIGURE P5–63

Given:

- State 1 (entering the valve)
  - $P_1 = 1.5$ MPa
  - It is a saturated liquid-vapor mixture at state 1.
- State 2 (exiting the valve)
  - $P_2 = 50$ kPa, $T_2 = 100$°C
  - At 50 kPa, $T_{\text{sat}} = 81.34$°C, so the steam is a superheated vapor at state 2. From the superheated water tables, we can find $h_2 = 2682.4$ kJ/kg.

Find:

- Quality of the steam at state 1, x_1

Writing the first law of thermodynamics for an open system:

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left[h_i + \frac{1}{2}v_i^2 + gz_i\right] - \sum \dot{m}_e \left[h_e + \frac{1}{2}v_e^2 + gz_e\right]$$

A throttling process is similar to a nozzle, except that the main role of a throttling process is to reduce the pressure of the fluid, not to increase its velocity. Therefore, we can simplify the first law equation as follows:

- No work is done by the system ($\dot{W} = 0$), it only reduces the pressure of the fluid
- No heat transfer ($\dot{Q} = 0$)
- Steady state ($\frac{dE}{dt} = 0$), because there is no accumulation of energy in the system over time
- Neglect potential energy changes ($gz_i \approx gz_e$)
- Neglect kinetic energy changes ($\frac{1}{2}v_i^2 \approx \frac{1}{2}v_e^2$)
- By conservation of mass, $\dot{m}_1 = \dot{m}_2$

Thus, the first law simplifies to:

$$0 = h_1 - h_2 \quad \implies \quad h_1 = h_2$$

<u>Note:</u> In a throttling process, the enthalpy of the fluid remains constant!

Therefore, using $h_1 = h_2$, we can find the quality of the steam at state 1:

$$x = \frac{h_1 - h_f}{h_{fg}} = \frac{[2682.4 \ - 844.55\,\text{kJ/kg}]}{[1946.4\,\text{kJ/kg}]} = \boxed{0.944}$$

5-73

5–73 Hot and cold streams of a fluid are mixed in a rigid mixing chamber. The hot fluid flows into the chamber at a mass flow rate of 5 kg/s with an energy in the amount of 150 kJ/kg. The cold fluid flows into the chamber with a mass flow rate of 15 kg/s and carries energy in the amount of 50 kJ/kg. There is heat transfer to the surroundings from the mixing chamber in the amount of 5.5 kW. The mixing chamber operates in a steady-flow manner and does not gain or lose energy or mass with time. Determine the energy carried from the mixing chamber by the fluid mixture per unit mass of fluid, in kJ/kg.

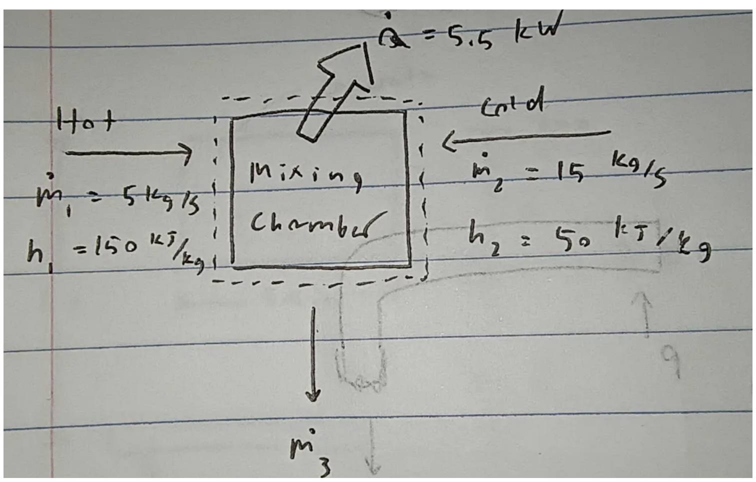
Given:

- State 1 (hot fluid entering the mixing chamber)
  - $\dot{m}_1 = 5$ kg/s, $h_1 = 1500$ kJ/kg
- State 2 (cold fluid entering the mixing chamber)
  - $\dot{m}_2 = 15$ kg/s, $h_2 = 50$ kJ/kg
- State 3 (mixture exiting the mixing chamber)
  - $\dot{m}_3 = \dot{m}_1 + \dot{m}_2$
- Heat transfer *from the system* to the surroundings, $\dot{Q} = 5.5$ kW

Find:

- Specific enthalpy of the mixture at state 3, h_3 [kJ/kg]

The system operates in a **steady-flow** manner, this means that there is no accumulation of energy or mass in the system over time. This is why we know there must be a $\dot{m}_3$ exiting the system:



Writing the first law of thermodynamics for an open system:

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left[h_i + \frac{1}{2}v_i^2 + gz_i \right] - \sum \dot{m}_e \left[h_e + \frac{1}{2}v_e^2 + gz_e \right]$$

We can simplify the first law equation as follows:

- No work is done by the system ($\dot{W} = 0$)
- Steady state ($\frac{dE}{dt} = 0$)
- Neglect potential energy changes ($gz_i \approx gz_e$)

- Neglect kinetic energy changes ($\frac{1}{2}v_i^2 \approx \frac{1}{2}v_e^2$)

Thus, the first law simplifies to:

$$0 = \dot{Q} + [\dot{m}_1 h_1 + \dot{m}_2 h_2] - \dot{m}_3 h_3$$

Substituting $\dot{m}_3 = \dot{m}_1 + \dot{m}_2$ and rewriting:

$$h_3 = \frac{\dot{Q} + \dot{m}_1 h_1 + \dot{m}_2 h_2}{\dot{m}_1 + \dot{m}_2}$$

<u>Note:</u> Since $\dot{Q}$ is heat transfer *from the system* to the surroundings, it is a negative value by sign convention.

$$h_3 = \frac{[-5.5\,\text{kW} \cdot \frac{1\,\text{kJ/s}}{1\,\text{kW}}] + [5\,\text{kg/s}][150\,\text{kJ/kg}] + [15\,\text{kg/s}][50\,\text{kJ/kg}]}{[5 + 15\,\text{kg/s}]} = \boxed{74.725\,\text{kJ/kg}}$$

5-78E

5–78E Steam is to be condensed on the shell side of a heat exchanger at 75°F. Cooling water enters the tubes at 50°F at a rate of 45 lbm/s and leaves at 65°F. Assuming the heat exchanger to be well-insulated, determine the rate of heat transfer in the heat exchanger and the rate of condensation of the steam.

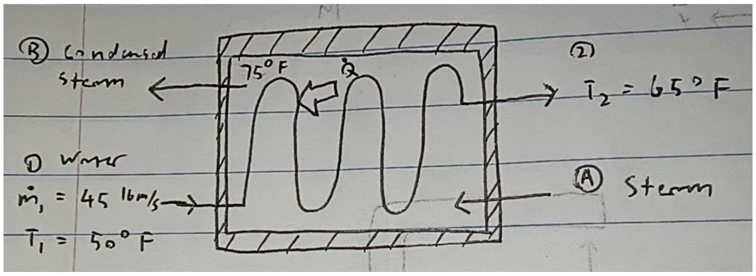
Given:

- State 1 (water entering the tube side)
  - $\dot{m}_1 = 45$ lbm/s, $T_1 = 50$°F
- State 2 (water exiting the tube side)
  - $T_2 = 65$°F
- State A (steam entering the shell side)
  - Assuming that the steam is a saturated vapor
- State B (condensed steam exiting the shell side)
  - Assuming that the condensed steam is a saturated liquid
- Condensation occurs in the shell side at $T = 75$°F, so the temperatures at states A and B are both 75°F.

  From the saturated water tables at $T = 75$°F, we can find $h_A = h_g = 1093.9$ Btu/lbm and $h_B = h_f = 43.07$ Btu/lbm.

Find:

- Rate of heat transfer from the steam to the water, $\dot{Q}$ [Btu/s]
- Rate of condensation of the steam, or in other words, the mass flow rate of the condensed steam exiting $\dot{m}_B$ [lbm/s]



Control Volume: Tube
Writing the first law of thermodynamics for an open system (tube side) to solve for the rate of heat transfer $\dot{Q}$:

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left[h_i + \frac{1}{2}v_i^2 + gz_i\right] - \sum \dot{m}_e \left[h_e + \frac{1}{2}v_e^2 + gz_e\right]$$

We can simplify the first law equation as follows:

- No work is done by the system ($\dot{W} = 0$)
- Steady state ($\frac{dE}{dt} = 0$)
- Neglect potential energy changes ($gz_i \approx gz_e$)
- Neglect kinetic energy changes ($\frac{1}{2}v_i^2 \approx \frac{1}{2}v_e^2$)
- By conservation of mass, $\dot{m}_1 = \dot{m}_2$

Thus, the first law simplifies to:

$$0 = \dot{Q} + \dot{m}_1 h_1 - \dot{m}_2 h_2$$

$$\dot{Q} = \dot{m}_1(h_2 - h_1)$$

Like problem 5-28E, we can find the change in specific enthalpy either using the property tables (compressed liquid approximation) or using the specific heat at constant pressure (c_p). Note that while c_p is constant for ideal gases, it is not constant for liquids. But we can still use it as an approximation since the water remains in the liquid phase and *does not change phases.*

I will use the compressed liquid approximation method: the properties of the water is approx the same as the properties of a saturated liquid at the same temperature. From the saturated water tables, we can find $h_1 = h_f = 18.07$ Btu/lbm and $h_2 = h_f = 33.08$ Btu/lbm.

$$\dot{Q} = [45\,\text{lbm/s}][(33.08 - 18.07)\,\text{Btu/lbm}] = \boxed{675.45\,\text{Btu/s}}$$

Control Volume: Shell

The simplified first law equation for the shell side is the same as the tube side, except that the mass flow rates are $\dot{m}_A = \dot{m}_B$ and the specific enthalpies are h_A and h_B:

$$0 = \dot{Q} + \dot{m}_A h_A - \dot{m}_B h_B$$

$$\dot{Q} = \dot{m}_B (h_B - h_A)$$

To find the rate of condensation of the steam, we can rearrange to solve for $\dot{m}_B$:

$$\dot{m}_B = \frac{\dot{Q}}{h_B - h_A}$$

We plug in the following values:

- The rate of heat transfer $\dot{Q}$ found before is the same except now it is *negative* because it is leaving the shell side

- Since we assumed that the steam is a saturated vapor at state A and a saturated liquid at state B, we can use the values of h_A and h_B found at $T = 75°\text{F}$ from the saturated water tables. $h_A = h_g = 1093.9$ Btu/lbm and $h_B = h_f = 43.07$ Btu/lbm.

$$\dot{m}_B = \frac{[-675.45\,\text{Btu/s}]}{[43.07 - 1093.9\,\text{Btu/lbm}]} = \boxed{0.642\,\text{lbm/s}}$$

5-185

5–185 A tank with an internal volume of 1 m^3 contains air at 800 kPa and 25°C. A valve on the tank is opened, allowing air to escape, and the pressure inside quickly drops to 150 kPa, at which point the valve is closed. Assume there is negligible heat transfer from the tank to the air left in the tank.

(*a*) Using the approximation $h_e \approx \text{constant} = h_{e,\text{avg}} = 0.5(h_1 + h_2)$, calculate the mass withdrawn during the process.

(*b*) Consider the same process but broken into two parts. That is, consider an intermediate state at $P_2 = 400$ kPa, calculate the mass removed during the process from $P_1 = 800$ kPa to P_2 and then the mass removed during the process from P_2 to $P_3 = 150$ kPa, using the type of approximation used in part (*a*), and add the two to get the total mass removed.

(*c*) Calculate the mass removed if the variation of h_e is accounted for.

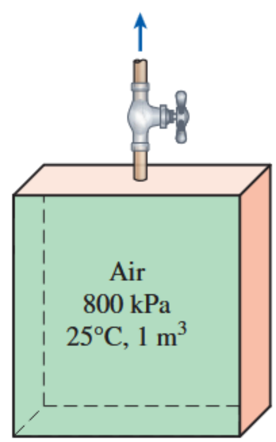


FIGURE P5–185

Given:

- State 1 (air in tank)
  - $\cancel{V}_1 = 1\,\text{m}^3$, $P_1 = 800$ kPa, $T_1 = 25°\text{C}$
- State 2 (intermediate state in tank)
  - $P_2 = 400$ kPa
- State 3 (air in tank)
  - $P_3 = 150$ kPa
- State e (air exiting the system)
  - Approximation $h_e \approx \text{constant} = h_{e,\text{ avg}} = \frac{1}{2}(h_1 + h_3)$
- No heat transfer ($\dot{Q} = 0$) to the air

Find:

- The mass exiting the system m_e [kg] using 3 different methods

Note that h_e is the specific enthalpy of the air exiting the system, and m_e is the total mass of the air that exits the system. Since the mass is changing over time, $\frac{dm}{dt} \neq 0$, so we cannot apply the steady state assumption.

Method (a):
As always starting with the first law of thermodynamics for an open system:

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left[h_i + \frac{1}{2}v_i^2 + gz_i\right] - \sum \dot{m}_e \left[h_e + \frac{1}{2}v_e^2 + gz_e\right]$$

We can simplify the first law equation as follows:

- No work is done by the system ($\dot{W} = 0$)
- No heat transfer ($\dot{Q} = 0$)
- Neglect potential energy changes ($gz_i \approx gz_e$)
- Neglect kinetic energy changes ($\frac{1}{2}v_i^2 \approx \frac{1}{2}v_e^2$)
- $\dot{m}_i = 0$ because there is no inlet

Thus, the first law simplifies to:

$$\frac{dE}{dt} = -\dot{m}_e h_e$$

We can integrate both sides over time to find the total mass exiting the system:

$$E_3 - E_1 = -m_e h_e$$

The change in the *total energy* of the system is equal to the change in the internal energy (U) of the air inside the tank since it is not flowing (so there is no flow work $h = u$).

$$E_3 - E_1 = m_3 u_3 - m_1 u_1$$

<u>Note:</u> We only use enthalpy h when there is flow work $P\cancel{v}$ (such as at the exit), but since there is no flow work at the inlet, we use internal energy u instead of enthalpy h for the states inside the tank.

$$\implies m_3 u_3 - m_1 u_1 = -m_e h_e$$

$$m_e = \frac{m_1 u_1 - m_3 u_3}{h_e}$$

Substituting in h_e:

$$m_e = 2\frac{m_1 u_1 - m_3 u_3}{(h_1 + h_3)}$$

To find the properties of the air at states 1 and 3, we can use the ideal gas properties of air table (A-17). But first we must find the temperature at state 3 using the i**sentropic relation**:

$$\boxed{\frac{T_3}{T_1} = \left(\frac{P_3}{P_1}\right)^{\frac{k-1}{k}}}$$

Where $k = c_p/c_{\cancel{v}}$ is the specific heat ratio.

We can assume that this is an isentropic process because there is no heat transfer and the process happens quick enough for there to be negligible frictional losses. Substituting in the values:

$$T_3 = T_1 \left(\frac{P_3}{P_1}\right)^{\frac{k-1}{k}}, \quad k = \frac{1.005}{0.718} = 1.399$$

$$T_3 = [298.15\,\text{K}] \left(\frac{[150\,\text{kPa}]}{[800\,\text{kPa}]}\right)^{\frac{1.399-1}{1.399}} = 184.966\,\text{K}$$

Knowing the temperatures at states 1 and 3, we can use the specific heat constants to find the specific internal energies and specific enthalpies at states 1 and 3:

- State 1 (assuming $h = 0$ and $u = 0$ at $T = 0$ K):

$$u_1 = c_v(T_1 - 0) = 0.718 \times (298.15 - 0) = 214.07\,\text{kJ/kg}$$

$$h_1 = c_p(T_1 - 0) = 1.005 \times (298.15 - 0) = 299.64\,\text{kJ/kg}$$

- State 3 (assuming $h = 0$ and $u = 0$ at $T = 0$ K):

$$u_3 = c_v(T_3 - 0) = 0.718 \times (184.966 - 0) = 132.80\,\text{kJ/kg}$$

$$h_3 = c_p(T_3 - 0) = 1.005 \times (184.966 - 0) = 185.89\,\text{kJ/kg}$$

Also, we can find the mass of the air at states 1 and 3 using the ideal gas law:

$$m = \frac{P\cancel{V}}{RT}$$

$$m_1 = \frac{[800 \times 10^3\,\mathrm{Pa}][1\,\mathrm{m}^3]}{[287\,\mathrm{J/kg \cdot K}][298.15\,\mathrm{K}]} = 9.349\,\mathrm{kg}$$

$$m_3 = \frac{[150 \times 10^3\,\mathrm{Pa}][1\,\mathrm{m}^3]}{[287\,\mathrm{J/kg \cdot K}][184.966\,\mathrm{K}]} = 2.825\,\mathrm{kg}$$

Finally, we can substitute all the values into the equation for m_e:

$$m_e = 2\frac{[9.349\,\mathrm{kg}][214.07\,\mathrm{kJ/kg}] - [2.825\,\mathrm{kg}][132.80\,\mathrm{kJ/kg}]}{[299.64 + 185.89\,\mathrm{kJ/kg}]} = \boxed{6.698\,\mathrm{kg}}$$

Method (b):
We do the same steps but with an intermediate state at state 2. We can find the temperature at state 2 using the isentropic relation:

$$T_2 = T_1 \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}} = [298.15\,\mathrm{K}] \left(\frac{[400\,\mathrm{kPa}]}{[800\,\mathrm{kPa}]}\right)^{\frac{1.399-1}{1.399}} = 244.66\,\mathrm{K}$$

Then we can find the specific internal energy, specific enthalpy, and mass at state 2:

$$u_2 = c_v(T_2 - 0) = 0.718 \times (244.66 - 0) = 175.66\,\mathrm{kJ/kg}$$

$$h_2 = c_p(T_2 - 0) = 1.005 \times (244.66 - 0) = 245.88\,\mathrm{kJ/kg}$$

$$m_2 = \frac{[400 \times 10^3\,\mathrm{Pa}][1\,\mathrm{m}^3]}{[287\,\mathrm{J/kg \cdot K}][244.66\,\mathrm{K}]} = 5.696\,\mathrm{kg}$$

Then we can apply the first law of thermodynamics to the system between states 1 and 2, and then between states 2 and 3, to find the mass exiting the system m_e:

$$m_{e_{1\to2}} = 2\frac{m_1u_1 - m_2u_2}{h_1 + h_2} = 2\frac{[9.349\,\mathrm{kg}][214.07\,\mathrm{kJ/kg}] - [5.696\,\mathrm{kg}][175.66\,\mathrm{kJ/kg}]}{[299.64 + 245.88\,\mathrm{kJ/kg}]} = 3.669\,\mathrm{kg}$$

$$m_{e_{2\to3}} = 2\frac{m_2u_2 - m_3u_3}{h_2 + h_3} = 2\frac{[5.696\,\mathrm{kg}][175.66\,\mathrm{kJ/kg}] - [2.825\,\mathrm{kg}][2.896\,\mathrm{kJ/kg}]}{[245.88 + 185.89\,\mathrm{kJ/kg}]} = 1.448\,\mathrm{kg}$$

$$m_e = m_{e_{1\to2}} + m_{e_{2\to3}} = 3.669 + 2.896 = \boxed{6.565\,\mathrm{kg}}$$

Method (c):
Now instead of treating h_e as a constant, we treat it as a variable. Going back to the differential form of the first law:

$$\frac{dU}{dt} = -\dot{m}_e h_e$$

$$\frac{d}{dt}(m_{\mathrm{sys}}u) = -\dot{m}_e h_e$$

$$\frac{d}{dt}(m_{\text{sys}}u) = m_{\text{sys}}\frac{du}{dt} + u\frac{dm_{\text{sys}}}{dt} = -\frac{dm_e}{dt}h_e$$

Note that by the conservation of mass, $\dot{m}_{\text{sys}} = -\dot{m}_e$, so we can rewrite:

$$\implies m_{\text{sys}}\frac{du}{dt} + u\frac{dm_{\text{sys}}}{dt} = \frac{dm_{\text{sys}}}{dt}h_e$$

Plug in for u using the ideal gas relation $u = c_v T$ and $h = c_p T$:

$$m_{\text{sys}}c_v\frac{dT}{dt} + c_v T\frac{dm_{\text{sys}}}{dt} = \frac{dm_{\text{sys}}}{dt}c_p T$$

"Multiply through" by dt and group differentials:

$$m_{\text{sys}}c_v\, dT = (c_p - c_v)T\, dm_{\text{sys}}$$

$$\frac{dT}{T} = \left(\frac{c_p}{c_v} - 1\right)\frac{dm_{\text{sys}}}{m_{\text{sys}}}$$

Integrate both sides:

$$\int_{T_1}^{T_3}\frac{dT}{T} = \left(\frac{c_p}{c_v} - 1\right)\int_{m_{\text{sys},1}}^{m_{\text{sys},3}}\frac{dm_{\text{sys}}}{m_{\text{sys}}}$$

$$\ln\left(\frac{T_3}{T_1}\right) = \left(\frac{c_p}{c_v} - 1\right)\ln\left(\frac{m_{\text{sys},3}}{m_{\text{sys},1}}\right)$$

From before, we know that $m_{\text{sys},1} = 9.349$ kg from the ideal gas law. Plugging in the values to solve for $m_{\text{sys},3}$:

$$\ln\left(\frac{T_3}{T_1}\right)\left(\frac{c_p}{c_v} - 1\right)^{-1} = \ln\left(\frac{m_{\text{sys},3}}{m_{\text{sys},1}}\right) = ln(m_{\text{sys},3}) - ln(m_{\text{sys},1})$$

$$ln(m_{\text{sys},3}) = ln(m_{\text{sys},1}) + \ln\left(\frac{T_3}{T_1}\right)\left(\frac{c_p}{c_v} - 1\right)^{-1}$$

$$m_{\text{sys},3} = m_{\text{sys},1}\cdot\exp\left[\ln\left(\frac{T_3}{T_1}\right)\left(\frac{c_p}{c_v} - 1\right)^{-1}\right]$$

$$m_{\text{sys},3} = 9.349\cdot\exp\left[\ln\left(\frac{184.966}{298.15}\right)\left(\frac{1.005}{0.718} - 1\right)^{-1}\right] = 2.831\,\text{kg}$$

FINALLY,

$$m_e = m_{\text{sys},1} - m_{\text{sys},3} = 9.349 - 2.831 = \boxed{6.518\,\text{kg}}$$

5-199

5–199 Steam is compressed by an adiabatic compressor from 0.2 MPa and 150°C to 0.8 MPa and 350°C at a rate of 1.30 kg/s. The power input to the compressor is

(*a*) 511 kW (*b*) 393 kW (*c*) 302 kW
(*d*) 717 kW (*e*) 901 kW

Given:

- State 1 (entering compressor)
  - $P_1 = 0.2$ MPa, $T_1 = 150°$C
- State 2 (exiting compressor)
  - $P_2 = 0.8$ MPa, $T_2 = 350°$C
- $\dot{m} = \dot{m}_1 = \dot{m}_2 = 1.3$ kg/s
- Adiabatic process (no heat transfer, $\dot{Q} = 0$) with STEAM

Find:

- Power input to the compressor, $\dot{W}_{\text{in}}$ [kW]

Writing the first law of thermodynamics for an open system (compressor):

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \sum \dot{m}_i \left[h_i + \frac{1}{2}v_i^2 + gz_i\right] - \sum \dot{m}_e \left[h_e + \frac{1}{2}v_e^2 + gz_e\right]$$

We can simplify the first law equation as follows:

- No heat transfer ($\dot{Q} = 0$)
- Steady state ($\frac{dE}{dt} = 0$)
- Neglect potential energy changes ($gz_i \approx gz_e$)
- Neglect kinetic energy changes ($\frac{1}{2}v_i^2 \approx \frac{1}{2}v_e^2$)

Thus, the first law simplifies to:

$$0 = -\dot{W} + \dot{m}_1 h_1 - \dot{m}_2 h_2 \implies \dot{W} = \dot{m}(h_1 - h_2)$$

Note that here $\dot{W}$ is the power done BY the system, so we substitute $\dot{W}_{\text{in}} = -\dot{W}$ to find the power input to the compressor:

$$\dot{W}_{\text{in}} = \dot{m}(h_2 - h_1)$$

Using the superheated water tables, we can find the specific enthalpies: $h_1 = 2769.1$ kJ/kg and $h_2 = 3162.2$ kJ/kg.

Substituting into the first law equation:

$$\dot{W}_{\text{in}} = [1.3\,\text{kg/s}][3162.2 - 2769.1\,\text{kJ/kg}] = \boxed{511.03\,\text{kW},\ (a)}$$